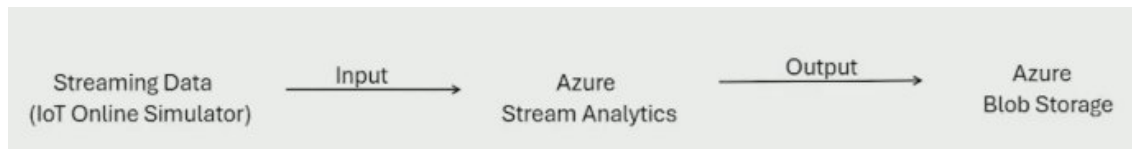
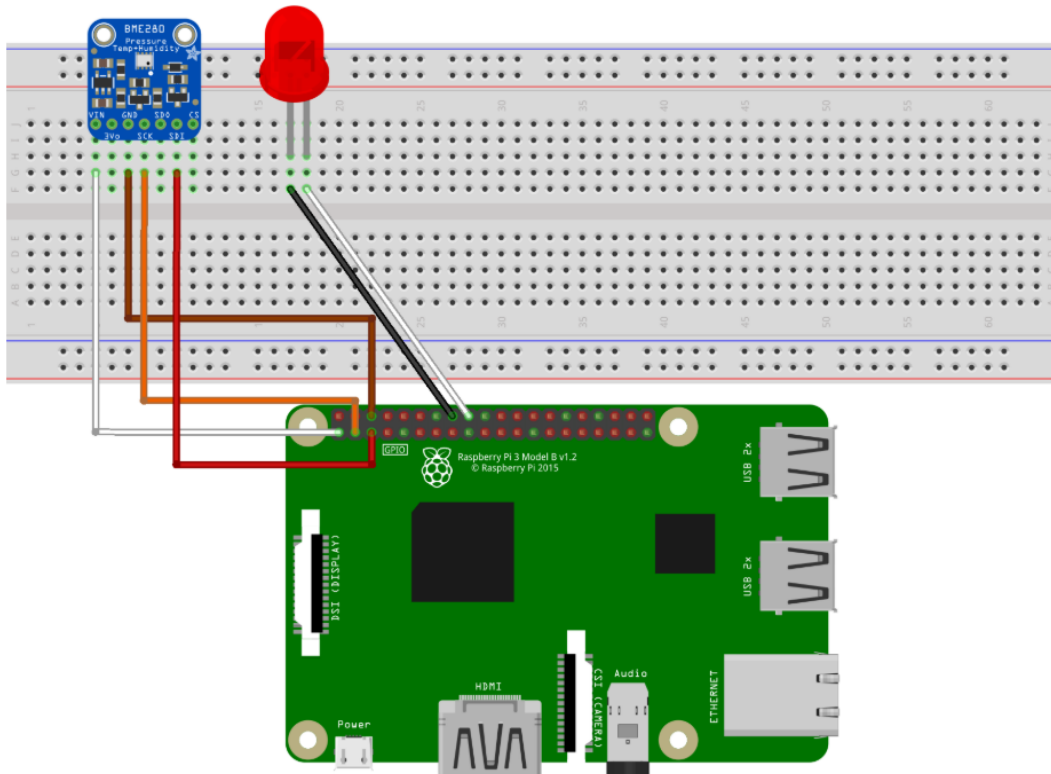


Streaming Flow:



Raspberry Pi Azure IoT Online Simulator:



IoT Hub Code:

```

1  /*
2  * IoT Hub Raspberry Pi NodeJS - Microsoft Sample Code - Copyright (c) 2017 - Licen
3  */
4  const wpi = require('wiring-pi');
5  const Client = require('azure-iot-device').Client;
6  const Message = require('azure-iot-device').Message;
7  const Protocol = require('azure-iot-device-mqtt').Mqtt;
8  const BME280 = require('bme280-sensor');
9
10 const BME280_OPTION = {
11     i2cBusNo: 1, // defaults to 1
12     i2cAddress: BME280.BME280_DEFAULT_I2C_ADDRESS() // defaults to 0x77
13 };
14
15 const connectionString = '[Your IoT hub device connection string]';
16 const LEDPin = 4;
17
18 var sendingMessage = false;
19 var messageId = 0;
20

```

Run Reset



Click `Run` button to run the sample code(when sample is running, code is read-only).
Click `Stop` button to stop the sample code running.
Click `Reset` to reset the code.We keep your changes to the editor even you refresh the page.
> ☐

Storage Account:

Home > stgstreams 1783947782351 | Overview > stgstreams | Containers

raw

Containers



+ Add Directory ↑ Upload ↻ Refresh 🗑 Delete 📄 Copy 📄 Paste 🔄 Rename ⚙ Acquire lease ⚙ Break lease 📄 Edit columns

Overview

🔧 Diagnose and solve problems

🔑 Access Control (IAM)

> Settings

raw

Authentication method: Access key ([Switch to Microsoft Entra user account](#))

Only show active objects

Showing all 0 items

☐	Name	Last modified	Access tier	Blob type	Size	Lease state
--------------------------	------	---------------	-------------	-----------	------	-------------

No items found

Stream Job Created:

stream-job-analytics
Stream Analytics job

[Run a query to find related resources](#) [What metrics are available for this resource](#) [Are there any outages impacting this resource](#)

Search

Start job Delete Move Refresh Share feedback 4 actions required

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Resource visualizer

Job topology

Settings

Developer tools

Monitoring

Automation

Help

Created

Essentials

Resource group (move)
rg-azure-july26

Location
Japan East

Status
Created

Subscription (move)
Pay-As-You-Go

Subscription ID
3c22336a-0420-4775-852c-127fd9da6056

Pricing plan
StandardV2 (manage)

Created
Monday, July 13, 2026 6:43 PM

Started

Output watermark

Cluster
Shared

Hosting environment
Cloud

Virtual Network
Disabled

JSON View

IoT Hub Creation:

IoTRaspberry
IoT Hub

[Summarize the properties of this IoT Hub.](#) [How do I troubleshoot issues with this IoT Hub?](#) [List access policies for this IoT Hub.](#)

Search

Move Delete Refresh Feedback

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Events

Resource visualizer

Device management

Devices

IoT Edge

Configurations + Deployments

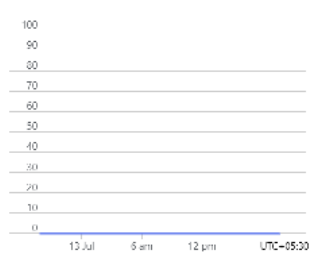
IoT Hub Usage

Messages used today: 0

Daily messages quota: 400000 ⓘ

IoT Devices: 0

Number of messages used



total number of messages used (Max): iot.hub... | 0

IoTRaspberry

IoTRaspberry

[Save](#) [Manage keys](#) [Message to Device](#) [Direct method](#) [Add Module Identity](#) [Device twin](#) [Refresh](#)

Device ID ⓘ

Primary key ⓘ

Secondary key ⓘ

Primary connection string ⓘ

Secondary connection string ⓘ

Tags (edit)

Enable connection to IoT Hub ⓘ

Parent device ⓘ

IoTRaspberry

.....

.....

.....

.....

No tags

☒ Enable ☐ Disable

No parent device

Registered with IoT Device to Raspberry Pi:

```
1  /*
2  * IoT Hub Raspberry Pi NodeJS - Microsoft Sample Code - Copyright (c) 2017 - Licen
3  */
4  const wpi = require('wiring-pi');
5  const Client = require('azure-iot-device').Client;
6  const Message = require('azure-iot-device').Message;
7  const Protocol = require('azure-iot-device-mqtt').Mqtt;
8  const BME280 = require('bme280-sensor');
9
10 const BME280_OPTION = {
11   i2cBusNo: 1, // defaults to 1
12   i2cAddress: BME280.BME280_DEFAULT_I2C_ADDRESS() // defaults to 0x77
13 };
14
15 const connectionString = 'HostName=IoTRaspberry.azure-devices.net;DeviceId=IoIRasp
16 const LEDPin = 4;
17
18 var sendingMessage = false;
19 var messageId = 0;
20
```

Run Reset

Click `Run` button to run the sample code(When sample is running, code is read-only).
Click `Stop` button to stop the sample code running.
Click `Reset` to reset the code.We keep your changes to the editor even you refresh the page.
> ☐

Input:

Input details



IoTRaspberry-input



Test



Delete



Open IoT Hub

Subscription

Pay-As-You-Go (3c22336a-0420-4775-852c-127fd9da6b56) ▾

IoT Hub * ⓘ

IoTRaspberry ▾

Consumer group * ⓘ

\$Default ▾

Shared access policy name * ⓘ

iothubowner ▾

Shared access policy key ⓘ

.....

Endpoint ⓘ

Messaging ▾

Save

Output:

Output details



IoTRaspberry-output



Test



Delete



Open Blob storage/ADLS Gen2

Subscription

Pay-As-You-Go (3c22336a-0420-4775-852c-127fd9da6b56) ▾

Storage account *

stgstreams ▾

Container * ⓘ



Create new



Use existing

raw ▾

Authentication mode

Connection string ▾

Storage account key ⓘ

.....

Save

Query:

[▶ Start job](#) [🔗 Open in VS Code](#) [🔗 Diagnostic settings](#) [🔄 Refresh](#) [📖 Query language docs ▾](#) [😊 Share feedback](#) [🎓 Tutorial](#) [🟢 Job ready to start](#)

✓ Inputs (1) +

IoTRaspberry input ...

✓ Outputs (1) +

IoTRaspberry output ...

✓ Functions (0) +

▶ Test query [📄 Save query](#) [✕ Discard changes](#)

```
1 select * into [IoTRaspberry-output]
2 from [IoTRaspberry-input]
3 having temperature > 25
```

Streams Running:

[❏ Stop job](#) [🔗 Open in VS Code](#) [🔗 Diagnostic settings](#) [🔄 Refresh](#) [📖 Query language docs ▾](#) [😊 Share feedback](#) [🎓 Tutorial](#)

ⓘ Query can't be edited while a job is running. You can stop the job to edit the query.

Data Started Generating:

```
1  /*
2  * IoT Hub Raspberry Pi NodeJS - Microsoft Sample Code - Copyright (c) 2017 - Licen
3  */
4  const wpi = require('wiring-pi');
5  const Client = require('azure-iot-device').Client;
6  const Message = require('azure-iot-device').Message;
7  const Protocol = require('azure-iot-device-mqtt').Mqtt;
8  const BME280 = require('bme280-sensor');
9
10 const BME280_OPTION = {
11   i2cBusNo: 1, // defaults to 1
12   i2cAddress: BME280.BME280_DEFAULT_I2C_ADDRESS() // defaults to 0x77
13 };
14
15 const connectionString = 'HostName=IoTRaspberry.azure-devices.net;DeviceId=IoIRasp
16 const LEDPin = 4;
17
18 var sendingMessage = false;
19 var messageId = 0;
20
```

Stop Reset

>
Message sent to Azure IoT Hub
>
Sending message: {"messageId":12,"deviceId":"Raspberry Pi Web Client","temperature":29.6382
>
Message sent to Azure IoT Hub
>
Sending message: {"messageId":13,"deviceId":"Raspberry Pi Web Client","temperature":25.4249
>
Message sent to Azure IoT Hub
> □

Data Received:

Test query Save query Discard changes

```
1 select * into [IoTRaspberry-output]
2 from [IoTRaspberry-input]
3 having temperature > 25
```

Input preview Test results Job simulation (preview)

Showing sample events from 'IoTRaspberry-input'.

Table Raw Refresh Select time range

messageId	deviceId	temperature	humidity	Event
<i>bigint</i>	<i>string</i>	<i>float</i>	<i>float</i>	<i>datetime</i>
1	Raspberry Pi Web Client	21.357435456677873	75.33386689600046	2026-
2	Raspberry Pi Web Client	20.814503732219237	67.35803444759293	2026-

Output based on Query i.e. filter > 25;

Input preview **Test results** Job simulation (preview)

Download results

messageId <i>bigint</i>	deviceId <i>string</i>	temperature <i>float</i>	humidity <i>float</i>	Event <i>datetime</i>
3	Raspberry Pi Web Client	27.476191522069072	73.55828682716546	2026-
4	Raspberry Pi Web Client	30.529290844438645	63.49997806835882	2026-

+ Add Directory ↑ Upload ↻ Refresh 🗑 Delete 📄 Copy 📄 Paste 🏷 Rename ⚙ Acquire lease ⚙ Break lease 🛠 Edit columns

raw

Authentication method: Access key (Switch to Microsoft Entra user account)

🔍 Search blobs by prefix (case sensitive)

Only show active objects

Showing all 1 items

Name	Last modified	Access tier	Blob type	Size	Lease state
📄 0_4e0f664944064...	13/07/2026, 20:13:51	Hot (Inferred)	Block blob	15.33 KiB	Available

Home > stgstreams_

raw

Container

🔍 Search

Overview

🔧 Diagnose and solve

🔑 Access Control (IAM)

> Settings

0_4e0f664944064928850bbd51af18bea2_1.json

Blob

📄 Save ✕ Discard ⬇ Download ↻ Refresh 🗑 Delete

Overview Versions **Edit** Generate SAS

```
1 {"messageId":3,"deviceId":"Raspberry Pi Web Client","temperature":27.476191522069072,"humidity":73.55828682716546,"EventProcessedUtcT":2026-07-13T20:13:51.0000000Z}
2 {"messageId":4,"deviceId":"Raspberry Pi Web Client","temperature":30.529290844438645,"humidity":63.49997806835882,"EventProcessedUtcT":2026-07-13T20:13:51.0000000Z}
3 {"messageId":7,"deviceId":"Raspberry Pi Web Client","temperature":26.72382130554788,"humidity":79.1421250258594,"EventProcessedUtcT":2026-07-13T20:13:51.0000000Z}
4 {"messageId":8,"deviceId":"Raspberry Pi Web Client","temperature":29.590462005966522,"humidity":63.124302928010174,"EventProcessedUtcT":2026-07-13T20:13:51.0000000Z}
5 {"messageId":9,"deviceId":"Raspberry Pi Web Client","temperature":27.822921587286544,"humidity":67.48285724978096,"EventProcessedUtcT":2026-07-13T20:13:51.0000000Z}
6 {"messageId":10,"deviceId":"Raspberry Pi Web Client","temperature":27.33075949656658,"humidity":75.16259228185794,"EventProcessedUtcT":2026-07-13T20:13:51.0000000Z}
7 {"messageId":11,"deviceId":"Raspberry Pi Web Client","temperature":29.61390487538672,"humidity":69.58486242576976,"EventProcessedUtcT":2026-07-13T20:13:51.0000000Z}
8 {"messageId":12,"deviceId":"Raspberry Pi Web Client","temperature":29.638291856875487,"humidity":76.89649895965307,"EventProcessedUtcT":2026-07-13T20:13:51.0000000Z}
9 {"messageId":13,"deviceId":"Raspberry Pi Web Client","temperature":25.424937671412106,"humidity":74.8291362091964,"EventProcessedUtcT":2026-07-13T20:13:51.0000000Z}
10 {"messageId":19,"deviceId":"Raspberry Pi Web Client","temperature":26.507484454331617,"humidity":72.35541572522988,"EventProcessedUtcT":2026-07-13T20:13:51.0000000Z}
11 {"messageId":20,"deviceId":"Raspberry Pi Web Client","temperature":29.47122395894863,"humidity":73.28291891408114,"EventProcessedUtcT":2026-07-13T20:13:51.0000000Z}
12 {"messageId":21,"deviceId":"Raspberry Pi Web Client","temperature":27.315742850453636,"humidity":73.03989252032673,"EventProcessedUtcT":2026-07-13T20:13:51.0000000Z}
13 {"messageId":22,"deviceId":"Raspberry Pi Web Client","temperature":27.623487572457,"humidity":64.65281732360047,"EventProcessedUtcT":2026-07-13T20:13:51.0000000Z}
14 {"messageId":23,"deviceId":"Raspberry Pi Web Client","temperature":25.59776609683015,"humidity":65.48454224011525,"EventProcessedUtcT":2026-07-13T20:13:51.0000000Z}
```

Json

Preview